

Case_ID	Temperature_C	Pressure_bar	Vibration_mm_s	Machine_Speed_RPM	Humidity_%	Material_Hardness_HRC	Observed_Defect	Root_Cause	Corrective_Action
C001	196	81	2.7	1415	41	53	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C002	237	112	5	1430	42	54	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C003	213	98	3.4	1445	43	55	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C004	224	104	2	1460	44	56	No Defect	Process within limits	No action required
C005	255	125	5.9	1475	45	57	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C006	201	81	2.5	1490	46	58	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C007	242	112	5.3	1505	47	59	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C008	218	98	3.2	1520	48	52	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C009	229	104	2.2	1535	49	53	No Defect	Process within limits	No action required
C010	260	120	5.7	1550	50	54	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C011	196	81	2.7	1565	51	55	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C012	237	112	5	1580	52	56	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C013	213	98	3.4	1595	53	57	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C014	224	104	2	1610	54	58	No Defect	Process within limits	No action required
C015	250	125	5.5	1625	55	59	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C016	201	81	2.5	1640	56	52	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C017	242	112	5.3	1655	57	53	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C018	218	98	3.2	1670	58	54	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C019	229	104	2.2	1685	59	55	No Defect	Process within limits	No action required
C020	255	120	5.9	1400	40	56	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C021	196	81	2.7	1415	41	57	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C022	237	112	5	1430	42	58	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C023	213	98	3.4	1445	43	59	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C024	224	104	2	1460	44	52	No Defect	Process within limits	No action required
C025	260	125	5.7	1475	45	53	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C026	201	81	2.5	1490	46	54	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C027	242	112	5.3	1505	47	55	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C028	218	98	3.2	1520	48	56	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C029	229	104	2.2	1535	49	57	No Defect	Process within limits	No action required
C030	250	120	5.5	1550	50	58	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C031	196	81	2.7	1565	51	59	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C032	237	112	5	1580	52	52	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C033	213	98	3.4	1595	53	53	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C034	224	104	2	1610	54	54	No Defect	Process within limits	No action required
C035	255	125	5.9	1625	55	55	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C036	201	81	2.5	1640	56	56	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C037	242	112	5.3	1655	57	57	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C038	218	98	3.2	1670	58	58	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C039	229	104	2.2	1685	59	59	No Defect	Process within limits	No action required
C040	260	120	5.7	1400	40	52	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C041	196	81	2.7	1415	41	53	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C042	237	112	5	1430	42	54	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C043	213	98	3.4	1445	43	55	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C044	224	104	2	1460	44	56	No Defect	Process within limits	No action required
C045	250	125	5.5	1475	45	57	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C046	201	81	2.5	1490	46	58	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C047	242	112	5.3	1505	47	59	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C048	218	98	3.2	1520	48	52	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C049	229	104	2.2	1535	49	53	No Defect	Process within limits	No action required
C050	255	120	5.9	1550	50	54	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C051	196	81	2.7	1565	51	55	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C052	237	112	5	1580	52	56	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C053	213	98	3.4	1595	53	57	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C054	224	104	2	1610	54	58	No Defect	Process within limits	No action required
C055	260	125	5.7	1625	55	59	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C056	201	81	2.5	1640	56	52	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C057	242	112	5.3	1655	57	53	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C058	218	98	3.2	1670	58	54	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C059	229	104	2.2	1685	59	55	No Defect	Process within limits	No action required
C060	250	120	5.5	1400	40	56	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C061	196	81	2.7	1415	41	57	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C062	237	112	5	1430	42	58	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C063	213	98	3.4	1445	43	59	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C064	224	104	2	1460	44	52	No Defect	Process within limits	No action required
C065	255	125	5.9	1475	45	53	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C066	201	81	2.5	1490	46	54	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C067	242	112	5.3	1505	47	55	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C068	218	98	3.2	1520	48	56	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C069	229	104	2.2	1535	49	57	No Defect	Process within limits	No action required
C070	260	120	5.7	1550	50	58	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C071	196	81	2.7	1565	51	59	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C072	237	112	5	1580	52	52	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C073	213	98	3.4	1595	53	53	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C074	224	104	2	1610	54	54	No Defect	Process within limits	No action required
C075	250	125	5.5	1625	55	55	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate
C076	201	81	2.5	1640	56	56	Dimensional Error	Insufficient pressure	Increase pressure by 10 bar and verify calibration
C077	242	112	5.3	1655	57	57	Surface Roughness	Excessive vibration	Inspect spindle and reduce vibration
C078	218	98	3.2	1670	58	58	Poor Finish	Tool wear	Replace cutting tool and verify speed settings
C079	229	104	2.2	1685	59	59	No Defect	Process within limits	No action required
C080	255	120	5.9	1400	40	52	Surface Crack	Excessive temperature	Reduce heater setting by 10°C and increase cooling rate